AP PHYSICS C

yeah

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0.1 Foreword for anybody else using this

As of writing this, I've already taken AP Physics 1 and 2 and Calc BC, so I'll skim over those stuff. I steal the images from google. If this violates copyright somehow, please email me and I'll remove it.

1 Kinematics

1.1 Scalars and Vectors

Scalars are numbers. Vectors are vectors. Represent as $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$.

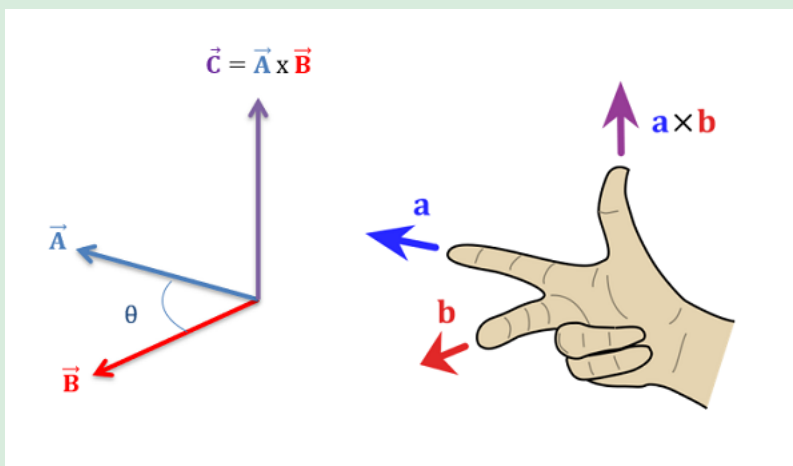
Dot product of two vectors is the length of one of the vectors projected onto the other multiplied by the length of said other vector. Theta = min angle between two vectors.

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

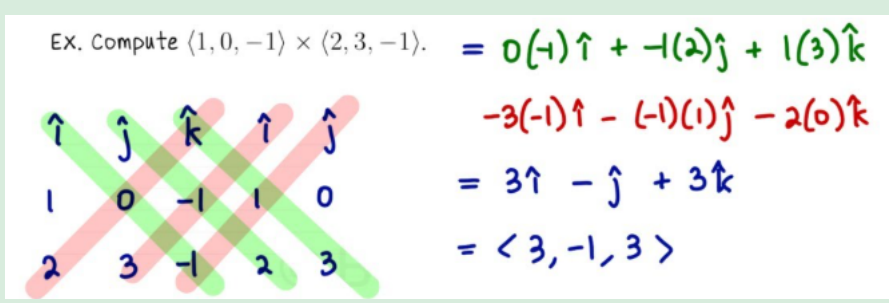
1.1.1 Cross Product

Operation that can be applied to two **three-dimensional** vectors to generate a third vector perpendicular to both vectors. Use the **right hand rule** to deterimine which direction the third vector will go.

Principle 1.1 (Right Hand Rule)
To find the geometrical direction of $\vec{a} \times \vec{b}$, hold your right hand out and point your *index* finger in the direction of $\vec{a}$, and your *middle* in the direction of $\vec{b}$. Your thumb will point in the direction of $\vec{a} \times \vec{b}$.



Principle 1.2 (Calculating the Cross Product)
To calculate the cross product between $\vec{A}$ and $\vec{B}$:



Note the following:

1. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$ where theta is the minimum angle betwen the two vectors. Also equal to the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.
2. $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. This property is known as the **anti-commutativity of the cross product**.
3. $\vec{a} \times \vec{a} = \vec{0}$ $(\theta = 0)$

1.2 Displacement, Velocity, and Acceleration

$$\vec{v} = \frac{\mathrm{d}\vec{s}}{\mathrm{d}t}, a = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\vec{s}}{\mathrm{d}t^2} \implies \Delta\vec{s} = \int \vec{v}(t)dt, \Delta\vec{v} = \int \vec{a}(t)dt$$

Derivatives and intergrals of vector-valued functions can be done component wise.

$$\vec{s} = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k} \implies \frac{\mathrm{d}\vec{s}}{\mathrm{d}t} = x'(t)\hat{\imath} + y'(t)\hat{\jmath} + z'(t)\hat{k}$$

$$\vec{s} = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k} \implies \int_{t_i}^{t_f} \vec{s}dt = \int_{t_i}^{t_f} x(t)dt + \int_{t_i}^{t_f} y(t)dt + \int_{t_i}^{t_f} z(t)dt$$

For constant one-dimensional acceleration, kinematic equations apply:

$$\Delta v = \int adt \implies v_f = v_i + at$$

$$\Delta s = \int v(t)dt = \int (v_i + at)dt \implies \Delta s = v_i t + \frac{1}{2}at^2$$

$$v_f = v_i + at \implies v_f^2 = v_i^2 + 2v_i at + a^2t^2 \implies v_f^2 = v_i^2 + a(2v_i t + at^2) \implies v_f^2 = v_i^2 + 2a\Delta x$$

Falling object experience constant acceleration: $a = g \approx 9.81\mathrm{m\,s^{-2}} \approx 10\mathrm{m\,s^{-2}}$. Projectiles experience constant vertical accleration of gravity, but no horizontal acceleration.

1.3 Representing Motion

Distance, velocity, and acceleration vs. time graphs exist. Yipee!

1.4 Reference Frames and Relative Motion

Let $\vec{v}_{AB}$ = velocity of A relative to B.

$$\vec{v}_{AB} = -\vec{v}_{BA}$$

$$\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}$$

These are also known as **Galilean Transformations.** Note these equations will break down near the speed of light due to relativity shit.

Example 1.3
The president's airplane flies at $250\mathrm{m\,s^{-1}}$ to the east with respect to the air. The air is moving at $35\mathrm{m\,s^{-1}}$ to the north with respect to the ground. Find the velocity of the president's airplane with respect to the ground.

Answer 1

Let $\hat{\imath}$ = north and $\hat{\jmath}$ = east.

$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG} = (250\hat{\jmath} + 35\hat{\imath})$$

$$|\vec{v}_{PG}| = \sqrt{250^2 + 35^2} \approx 252.4\,\mathrm{m\,s^{-1}}$$

$$\theta_{PG} = \arctan\frac{35}{250} \approx 7.97^\circ \text{ relative to East}$$

1.5 Motion in Two or Three Dimensions

everything that we've talked about already considers three dimensions :3

2 Force and Translational Dynamics

2.1 Systems and Center of Mass

Recall that the location of a system's center of mass along a given axis can be calculated as follows:

$$\vec{x}_{cm} = \frac{\sum m_i \vec{x}_i}{\sum m_i}$$

This equation can be extended for non-uniform solids by considering differential masses $\mathrm{d}m$:

$$\vec{r}_{cm} = \frac{\int \vec{r}\,\mathrm{d}m}{\int \mathrm{d}m}$$

One way to calculate $\mathrm{d}m$ is through **linear mass density**:

$$\lambda = \frac{\mathrm{d}m}{\mathrm{d}l}$$

Area and volume mass density exists as well, represented by σ and ρ respectively.

Example 2.1
Suppose you have a bar of length L which has mass density function $m(r) = cr^2$. Find the center of mass in terms of L, c, and fundamental constants.

Answer 2

$$\frac{\mathrm{d}m}{\mathrm{d}r} = 2cr \implies \mathrm{d}m = 2cr\,\mathrm{d}r$$

$$\begin{aligned} r_{cm} &= \frac{\int r\,\mathrm{d}m}{\int \mathrm{d}m} \\ &= \frac{\int_0^L 2cr^2\,\mathrm{d}r}{\int_0^L 2cr\,\mathrm{d}r} \\ &= \frac{\frac{r^3}{3}\Big|_0^L}{\frac{r^2}{2}\Big|_0^L} \\ &= \frac{2}{3}\frac{L^3}{L^2} \\ &= \boxed{\frac{2}{3}L} \end{aligned}$$

Also recall that for force diagrams, systems can be modeled as a single point located at the system's center of mass.

2.2 Forces and Free-Body Diagrams

they exist

2.3 Newton's Third Law

$$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}}$$

Recall that tension is the macroscopic net of forces segments of a string, cable, chain or a similar system exertes on each other. The tension has negligible mass, does not stretch, and has equal tension across all points in the string.

2.4 Newton's First Law

It's called **translational equilibrium** if the net force on the system is zero.

2.5 Newton's Second Law

$$\vec{F}_{net} = m_{sys}\vec{a}_{sys}$$

2.6 Gravitational Force

Definition 2.2 (Newton's law of universal gravitation)

$$|\vec{F}_g| = G\frac{m_1 m_2}{r^2}$$

The gravitational force is always exerted along the line connecting the center of masses of two systems.

This force suggests that a mass sets up a **gravitational field** around itself: $|\vec{g}| = G\frac{M}{r^2}$. The word **weight** is only used when one of the objects is much more massive than the other (usually the Earth). **Apparent weight** is the normal force acting on an object, and is what is measured by a scale in an elevator. Near the surface of the earth, $g \approx 10\text{N}\,\text{kg}^{-1}$.

Newton's Shell Theorem states *the net gravitational force exerted on an object inside a thin spherical hell is zero* AND *the net gravitational force exerted on an object by a spherical shell is equiavlent to treating the shell as a point mass.*

In practice, the first half of Newton's Shell Theorem suggests that **within a uniform sphere, only the mass of a sphere within a radius equal to the object's distance from the center** affects it. All outer layers can be considered as differential spherical shells, and by Newton's Shell Theorem exert no gravitational force. Thus,

$$m_{partial} = \rho\frac{4}{3}\pi(r_{partial})^3 \text{ (this equation is NOT on the sheet)}$$

Using the second half of Newton's Shell theorem and the law of gravitation:

$$\begin{aligned} F_g &= \frac{Gm}{r_{partial}^2} * \rho\frac{4}{3}\pi(r_{partial})^3 \\ &= \frac{4}{3}Gm\rho\pi r_{partial} \\ &\implies F_g \propto r_{partial} \text{ (linear proportionality within the sphere!)} \end{aligned}$$

Example 2.3
Find the weight of a 1.0 kg mass located half way to Earth's center. Ignore density variation.

Answer 3

$$F_g \propto r_{partial} \implies F_g = \frac{mg}{2} = \frac{1.0 * 10}{2} = \boxed{5\text{N}}$$

2.7 Kinetic and Static Friction

Frictional force is exerted to oppose a change in motion on a rough surface, and is proportional to the magnitude of the normal force. The proportionality constant depends on whether it is moving (**kinetic**) or stationary (**static**). Kinetic friction is usually less than static friction.

2.8 Spring Forces

$$\vec{F}_{spr} = -k\Delta x$$

$$k_{\text{eq,parallel}} = \sum k_i$$

$$\frac{1}{k_{\text{eq,series}}} = \sum \frac{1}{k_i}$$

2.9 Resistive Forces

A resitive force is defined as a **velocity-dependent** force that opposes an object's velocity.

$$\vec{F}_{res} = -k\vec{v}$$

Setting up a second law equation with a resistive force will lead to a differential equation for velocity (recall $a = \frac{\mathrm{d}v}{\mathrm{d}t}$), which can be solved to lead to an exponential equation for velocity. Terminal velocity is defined as the maximum velocity reached under a resistive force.

Example 2.4
A skydiver of mass m drops from rest, and experiences air resistance $F_{air} = -bv$ Assuming they do not hit the ground, find an equation for their velocity as a function of time. State what this suggests about their terminal velocity.

Answer 4

$$
\begin{aligned}
F_{net} &= mg - F_{air} \\
m\frac{\mathrm{d}v}{\mathrm{d}t} &= mg - bv \\
\frac{m}{mg - bv}\,\mathrm{d}v &= 1\,\mathrm{d}t \\
\int \frac{m}{mg - bv}\,\mathrm{d}v &= \int \mathrm{d}t \\
-m \ln mg - bv &= t \\
mg - bv &= e^{-\frac{t}{m}} \\
&\implies \boxed{v = \frac{mg}{b}(1 - e^{-\frac{t}{m}})}
\end{aligned}
$$

As $t \to \infty$, $\boxed{v \to \frac{mg}{b}}$

2.10 Circular Motion

In order to maintain a cirular path of radius r, a particle needs to experience $a_c = \frac{v^2}{r}$ directed towards the center of the motion. This is known as **centripetal acceleration**. The acceleration along the path of travel is known as **tangential acceleration**. Any acceleration vector can be split into centripetal and tangential components.

Note that for a vertical, circular loop, an object requires a minimum speed to maintain circular motion as the gravitational force is the only force that causes centripetal acceleration.

$$v = \sqrt{gr}$$

3 Work and Power

3.1 Translational Kinetic Energy

$$\mathrm{KE} = \frac{1}{2}mv^2$$

Note different observers may observe different translational kinetic energies depending on if the observer is moving as well.

3.2 Work

Work is the amount of energy transferred in or out of a system by a force exerted by that system over a distance. Potential energies are always associated with only **conservative** (path-independent) forces. The most common *non-conservative* forces are air resistance and friction.

The non-calculus work formula $W = Fd\cos\theta$ can be extended to any path:

$$W = \int_a^b \vec{F}(r) \cdot \mathrm{d}\vec{r}$$

This formula suggests that ony the component of the force parallel to displacement can change a system's energy, and suggests that work is equal to the area under the curve of a graph of $F_{\parallel}(r)$.

The **work-energy theorem** states the change in an object's kinetic energy is equal to the sum of the work being done by all forces.

3.3 Potential Energy

The relationship between conervative forces exerted on a system and the system's potential energy is

$$\Delta U = -\int_a^b \vec{F}_{cf}(r) \cdot \mathrm{d}\vec{r}$$

($\vec{F}_{cf}$ = conservative forces) Or, alternatively (not on the equation sheet)

$$\Delta U = -W_{cf} \text{ (potential energy is converted into work)}$$

This definition can be used to prove the potential energy equations for springs and universal gravitation:

Proof of spring energy.

$$\begin{aligned}
\Delta U_{spr} &= -\int \vec{F}_{spr} \cdot \mathrm{d}\vec{r} \\
&= -\int -kx \,\mathrm{d}x \\
&\implies \boxed{\Delta U_{spr} = \frac{1}{2}kx^2}
\end{aligned}$$

□

Proof of universital gravitational energy.

$$\begin{aligned}\Delta U_g &= -\int \vec{F}_g \cdot \mathrm{d}\vec{r} \\ &= -\int G\frac{m_1 m_2}{r^2}\,\mathrm{d}r \\ &\implies \boxed{\Delta U_g = G\frac{m_1 m_2}{r}}\end{aligned}$$

□

The intergral definition of potential energy suggests that

$$F_{cf} = -\frac{\mathrm{d}U}{\mathrm{d}x}$$

This further suggests **stable equlibria** occur at local minima of the potential energy function (F_{cf} opposes change) while **unstable equilibria** occur at the maxima of the potential energy function (F_{cf} supports change)

3.4 Conservation of Energy

$$\mathrm{ME} = \mathrm{KE} + \mathrm{PE}$$

$$\Delta\mathrm{ME} = W_{nc} = \text{work transferred out of the system}$$

3.5 Power

$$P_{avg} = \frac{\Delta E_{transferred}}{\Delta t} = \frac{W}{\Delta t} \implies P_{inst} = \frac{\mathrm{d}W}{\mathrm{d}t}$$

If an object is moving at constant velocity $\vec{v}$ due to an external force $\vec{F}$, then the power transferred to the object is

$$P = \vec{F} \cdot \vec{v}$$

This usually applies to the force that a car exterts to drive at a given velocity.

4 Linear Momentum

5 Torque and Rotational Dyanmics

6 Energy and Momentum of Rotating Systems

7 Oscillations

8 Electric Charges, Fields, and Gauss's Law

8.1 Electric Charge and the Electric Force

Charge... exists. It's scalar and measured in SI units **Coulombs**.

Coulomb's Law describes the electrostatic force between two charged particles:

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2} = k\frac{|q_1 q_2|}{r^2}$$

ϵ_0 represents the **electric permittivity of free space** and has a constant value (units $\mathrm{C^2N^{-1}m^{-2}}$). Electric permittivity measures "the degree to which a material or medium is polarized in the presence of an electric field" and is determined by "the ease with which electrons can change configurations within a given material." Electric polarization can be modeled as the induced rearangement of electrons by an external electric field.

8.2 Conservation of Electric Charge and the Process of Charging

Like charges repel. Charges on system influence each other.

8.3 Electric Fields

Electric fields exist. $\vec{E} = \frac{\vec{F_e}}{q_e} = k\frac{q}{r^2}\hat{r}$

Note while in electrostatic equilibrium (no net movement of charges), the excess charge of a conductor is distributed on the surface of said conductor, and the electric field within said conductor is 0. Note that this is **not** true for insulators: charge is distributed through the surface AND interior, and the electric field within may be non-zero.

8.4 Electric Fields of Charge Distributions

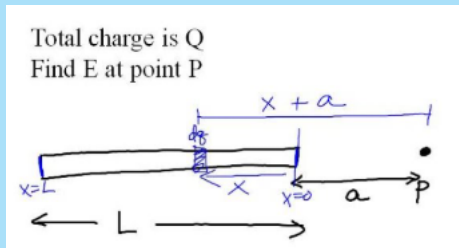
For continious charge distributions, you can consider tiny point charges $\mathrm{d}q$ and apply Coulomb's Law to each of them, then add their effects together. The fact that you can add them together is known as **the principle of superposition.** Mathematically:

$$\vec{E} = k\int \frac{\mathrm{d}q}{r^2}\hat{r}$$

You can typically directly solve for Q or express the integral in terms of $\mathrm{d}r$ using **linear charge density** $\lambda = \frac{\mathrm{d}Q}{\mathrm{d}L} \approx \frac{Q}{L}$, **surface charge density** $\sigma = \frac{Q}{A}$, or **volumetric charge density** $\rho = \frac{Q}{V}$

Example 8.1

Determine the electric field at point P.



Answer 5

$$\lambda \, \mathrm{d}r = \mathrm{d}Q$$

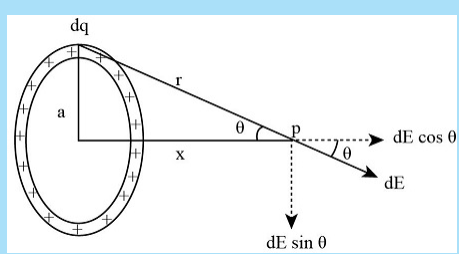
$$\begin{aligned}
\vec{E} &= k \int \frac{\mathrm{d}Q}{r^2} \hat{r} \\
&= k\hat{r} \int_a^{a+L} \frac{\lambda \, \mathrm{d}r}{r^2} \\
&= \lambda k \hat{r} \left(\frac{-1}{a+L} + \frac{1}{a} \right) \\
&= \lambda k (\hat{\imath}) \left(\frac{L}{a(a+L)} \right) \\
&= \boxed{\frac{Qk\hat{\imath}}{a(a+L)}}
\end{aligned}$$

Note when $a >> L$, it degenerates the the below, which suggests that it acts as a point charge at sufficently large distances.

$$\vec{E} \approx \frac{Qk\hat{\imath}}{a^2}$$

Example 8.2

Determine the electric field at P.



Answer 6
Note that "horizontal" $\mathrm{d}\vec{E}\sin\theta$ components cancel via symmetry. Thus, you only need to consider components proportional to $\cos\theta$.

$$\begin{aligned}\vec{E} &= k\int \frac{\mathrm{d}Q}{a^2+x^2}\hat{\imath}\cos\theta \\ &= k\hat{\imath}\int \frac{\mathrm{d}Q}{a^2+x^2} \times \frac{x}{\sqrt{a^2+x^2}} \\ &= \boxed{\frac{k\hat{\imath}Qx}{(a^2+x^2)^{1.5}}}\end{aligned}$$

8.5 Electric Flux

Flux describes the amount of a given quantity that passes through a given area. **Electric flux** Φ_E describes the amount of electric field that passes through a defined area. Note the area vector is defined to be outward from a closed surface.

$$\Phi_E = \vec{E}\cdot\vec{A}$$

For an arbitrary surface:

$$\Phi_E = \int \vec{E}\cdot\mathrm{d}\vec{A}$$

8.6 Gauss's Law

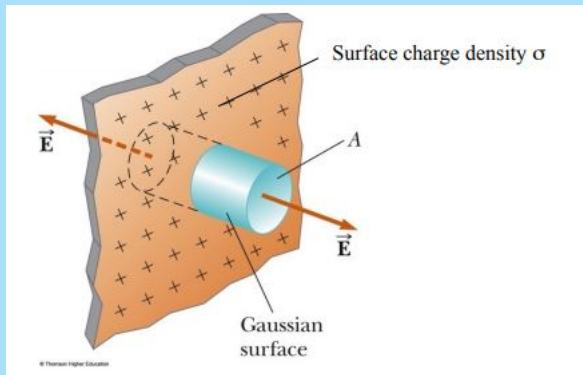
Definition 8.3 (Gauss's Law)
For an arbitrary **Gaussian Surface** (3d closed surface):

$$\Phi_E = \oint \vec{E}\cdot\mathrm{d}\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

Usually you can rig the surface such that the electric field is perpendicular or parallel to all components of the area, with a sphere or a cylinder.

Example 8.4

Determine the electric field surrounding an infinitely large, thin plane of positive changes with uniform surface charge density.



Answer 7

Set up the Gaussian surface as shown. The area vector is perpendicular to the electric field along the "wrapping" of the cylinder. The area vector is parallel to the electric field in the same direction for the two "caps" of the cylinder. Thus,

$$2EA = \frac{q_{enc}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} \implies E = \boxed{\frac{\sigma}{2\epsilon_0}}$$

Note for the case of a two "infinite" plane capacitor, we assume the field outside the capacitor cancels. Thus,

$$E_{capacitor} = \frac{\sigma}{\epsilon_0}$$

9 Electric Potential

9.1 Electric Potential Energy

EPE of a system = amount of work required by external force to bring point charges to current positions from infinitely far away. Total EPE = sum of EPE between each pair of charged particles.

$$U_E = k\frac{q_1 q_2}{r}$$

Proof of EPE formula. Bringing q_1 to q_2 from infinity is equivalent to bringing q_2 to q_1 from infinity. EPE gained by work done to bring one charge to the given distance away.

$$\begin{aligned}
\text{EPE} = W &= -\int \vec{F} \cdot \mathrm{d}\vec{r} \\
&= -\int_{-\infty}^{r} k\frac{q_1 q_2}{x^2}\,\mathrm{d}x \\
&= -kq_1q_2(-\frac{1}{r} - \frac{1}{\infty}) \\
&= \boxed{k\frac{q_1 q_2}{r}}
\end{aligned}$$

□

9.2 Electric Potential

Electric potential describes the *potential energy* per unit charge at a point in space. Note it is a scalar. The electric force is **conservative**: change in EPE due to position change does not depend on the path. Electric potentials caused by different charges add.

$$V = k\frac{q}{r} \Leftrightarrow V = k\int \frac{\mathrm{d}q}{r}$$

The electric potential difference between two points is the change in EPE between the two points:

$$\Delta V = \frac{\Delta U_e}{q} = -\int \vec{E} \cdot \mathrm{d}\vec{r} = -Ed$$

for perpendicular distance d. The integral form comes from the work form of EPE divided by q.

Note the integral form suggests the following:

$$\vec{E}_x = -\frac{\mathrm{d}V}{\mathrm{d}x}$$

Also note the definition of potential suggests positive charges want to go to low electric potential (loses EPE =¿ KE)

9.3 Conservation of Electric Energy

EPE is a type of energy, so it is conserved by the law of energy conservation.

10 Conductors and Capacitors

11 Electric Circuits

12 Magnetic Fields and Electromagnetism

13 Electromagnetic Induction